

Airline Delay Project

2025 implementation progress update | Data wrangling, initial cleaning, and first-pass EDA

Scope

The project uses all 12 monthly 2025 BTS Reporting Carrier On-Time Performance files and the supplied OurAirports metadata. The goal is to analyze operational delay patterns and later build a leakage-safe model that predicts severe arrival delays before departure.

What has been completed

7,001,619	12 / 12	28	100%
BTS flight rows audited	Monthly files validated	Consistent source columns	Airport-code match

- Raw BTS and airport data reorganized into a reproducible 2025-only project structure without modifying source contents.
- Master plan updated from 2023-2025 to one full 2025 calendar year, including a chronological within-year ML split.
- Chunked Python pipeline created for source manifesting, cleaning, validation, feature engineering, airport-reference standardization, and EDA.
- Reports, summary tables, figures, settings, column mapping, README instructions, and cleaning tests added.

Current project structure

airline-delay-project/
data/raw/bts/2025/ - 12 renamed monthly BTS files
data/raw/airports/ - airports.csv
config/ - settings and column mapping
src/ and scripts/ - reusable pipeline code
reports/ - quality report, cleaning log, EDA tables, figures

Current phase: first-pass cleaning and EDA are complete. The project has passed the requested 50% cleaning/EDA checkpoint.

Cleaning, validation, and data quality

The pipeline reads the full year in chunks, protecting the raw data while keeping the analysis reproducible.

Cleaning and transformation work completed

- Mapped BTS uppercase source columns to documented snake_case analysis names; parsed flight dates and numeric measures.
- Trimmed and standardized airline, airport, and cancellation codes. Created route, scheduled departure hour, departure-time band, weekend, season, delay category, and severe-delay features.
- Defined severe delay as ARR_DELAY greater than 60 minutes only for completed, non-diverted flights with a known arrival outcome.
- Flagged five impossible nonpositive scheduled-duration values and set them to null only in derived data; raw records remain untouched.
- Standardized the airport reference and documented a historical PBI to DJT alias found in the supplied airport metadata, producing full origin and destination code coverage.

Measured validation outcomes

Check	Outcome
Monthly source coverage	12 of 12 files present; each file dates only to its expected 2025 month
Schema and duplicate audit	One consistent 28-column schema; 0 exact and 0 business-key duplicates
Code and flag validation	0 invalid origin/destination codes; 0 invalid cancellation or diversion flags
Airport join	352 distinct BTS codes matched for origins and destinations (100% flight-row match)
Target eligibility	6,879,484 completed, non-diverted flights with known arrival delay

Important handling decisions

Cancelled and diverted flights keep their operational records but do not receive a severe-arrival-delay target. Missing post-flight fields and extreme delays are retained for analysis rather than imputed or removed automatically. Reported delay-cause fields are descriptive only and are excluded from the future pre-departure ML feature set.

Six records are flagged for contextual follow-up: five contain impossible scheduled elapsed durations and one completed flight has no arrival-delay value. They do not block the current EDA; their details are preserved in the flagged-row table.

Still pending in cleaning: contextual review of extreme delay events, finalized retained-column schema, and persisted cleaned flight partitions.

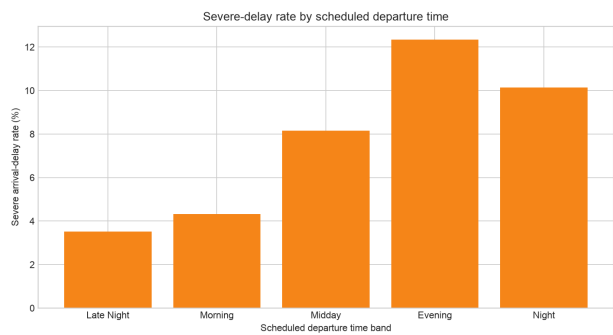
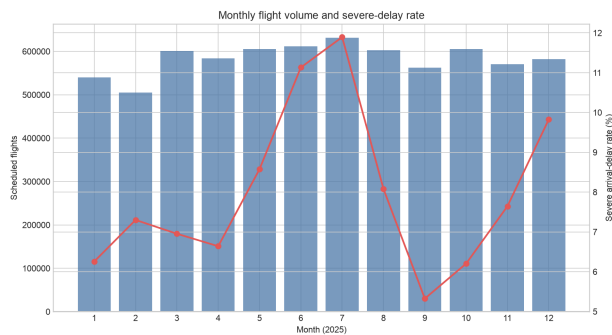
Initial EDA outcomes and next steps

All values below were calculated from the 2025 files, not assumed in advance.

8.5 min	-6 min	8.04%	1.47%
Average arrival delay	Median arrival delay	Severe-delay rate	Cancellation rate

Evidence-backed findings

- July had the highest monthly severe-delay rate: 11.89% across 612,811 eligible flights.
- Evening departures had the highest severe-delay rate: 12.34%, compared with the full-year rate of 8.04%.
- Among airlines with at least 10,000 flights, OH had the highest severe-delay rate at 12.12%.
- Among routes with at least 1,000 flights, ASE-DFW had the highest severe-delay rate at 19.15%.
- Late-aircraft delay accounted for 39.19% of reported delay-cause minutes, the largest category. This is descriptive only, not an ML predictor.



Next implementation sequence

1. Review the flagged rows and extreme-delay context. 2. Persist standardized flight partitions and complete the airport enrichment. 3. Create viva-friendly audit, cleaning, and EDA notebooks. 4. Load the cleaned data into SQLite. 5. Freeze the chronological 2025 train/validation/test split, then compare a baseline with Logistic Regression and a Decision Tree.

Limitation: one full year supports within-2025 analysis but cannot establish year-over-year trends or long-term generalization.